

Project Gutenberg language diversity over time

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Abstract

context for understanding the evolution of digital preservation efforts.

1 Introduction

This study analyses the linguistic diversity in Project Gutenberg's digital library collection over time, examining changes in both English and non-English books added to the Gutenberg Collection per year. It aims to answer what the difference is in the proportion of languages added to Project Gutenberg over time. This research analyses the works added year over year to the Gutenberg collection, showing the changes in addition rates of differing languages. The hypothesis is that over time, the proportion of English books will decrease compared to non-English books, reflecting an increase in internet availability and interest in preservation of historical books, and a decrease in available public domain books written in English.

2 Related Work

Several studies have examined linguistic diversity in digital libraries and Project Gutenberg specifically:

1. **whitt`diachronic`2018** explored multilingualism on the web and Project Gutenberg's early efforts at linguistic diversification, providing important historical

2. **gerlach`standardized`2020** analysed a curated Project Gutenberg database, revealing a strong bias toward Western languages, with English comprising 81% of all books. Their work established baseline metrics for understanding linguistic representation in digital libraries.

Project Gutenberg also knows about the disparity in languages in its collection. For example, on the frequently asked questions pages on the [gutenberg.org](https://www.gutenberg.org) website they explain how books are added, namely by volunteers. This makes it difficult for Project Gutenberg to directly change which books are added, as they can't dictate what these volunteers add.

3 Data

Pre-processing The data pre-processing was done by downloading the metadata of all Project Gutenberg books from the [gutenberg.org](https://www.gutenberg.org) website on 06-01-2025. This data can be downloaded in the form of a CSV file, which contains the relevant fields for this article: the date added and the language. Books that included multiple languages, either due to being a translation that includes the original text or being a translation dic-

Ywe	English	Other
1970-1980	9	0
1881-1990	1	0
1991-2000	1861	79
2001-2010	27375	5526
2011-2020	22767	6703
2021-2024	7970	2759

Table 1: Books added to Project Gutenberg per decade

tionary, are counted for both languages. The python script used for extracting the data will be included with the paper. Due to not being completed, and thus giving incomplete information, the year 2025 was excluded. As only the language codes were given with the metadata, language codes were manually linked to their respective language. This resulted in a matrix for book additions per year for every language, from the years 1971 to 2024, with 70 languages. The non-English books were all summed, creating the distribution graph shown in figure 1. Books added per decade are shown in table 1. The complete data matrix is shown in table 2 in the Appendix, with the total books per language shown in figure 2, also in the appendix.

4 Predicted Results

As seen in figure 1, the proportion of English to non-English books has significantly changed over time, particularly after the large increase in books added in 2004. But while English books experienced a sharp decline after this peak of over 6,000 additions, non-English books maintained a relatively stable

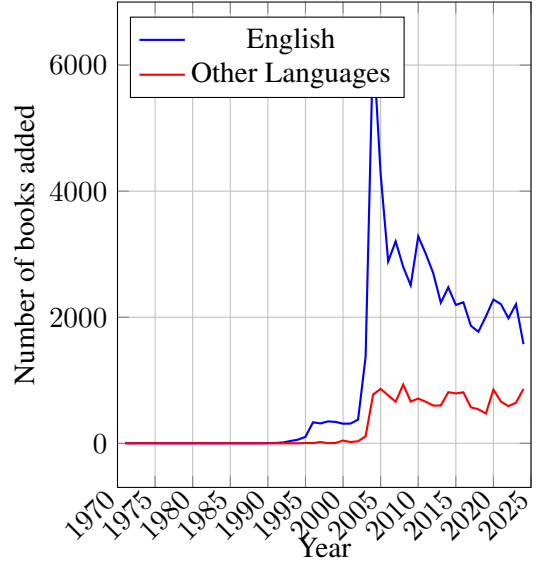


Figure 1: Number of books added to Project Gutenberg per year for English and non-English languages

addition rate of 600-800 books per year following 2004. Furthermore, English book additions have begun to steadily decline after the 2004, if inconsistently. If this pattern continues, the gap between English and non-English book addition rates will likely become narrower in the future.

Discussion The trend shown in the data strongly supports the hypothesis that the proportion of English books would decrease compared to non-English books. The decline in English book additions could be attributed to the fact that most English public domain books have already been entered, with remaining books becoming increasingly rare. The stability in non-English additions suggests growing international attention for digital preservation efforts. This could be due to several factors: increased global internet accessibility, greater awareness of Project Gutenberg internationally, and improved or more accesible optical character recognition

(OCR) technology for non-Latin scripts, making it easier to participate. Additionally, as many English books have already been digitized, volunteers might be increasingly focussing on preserving texts in other languages.

5 Conclusion

6 Appendix

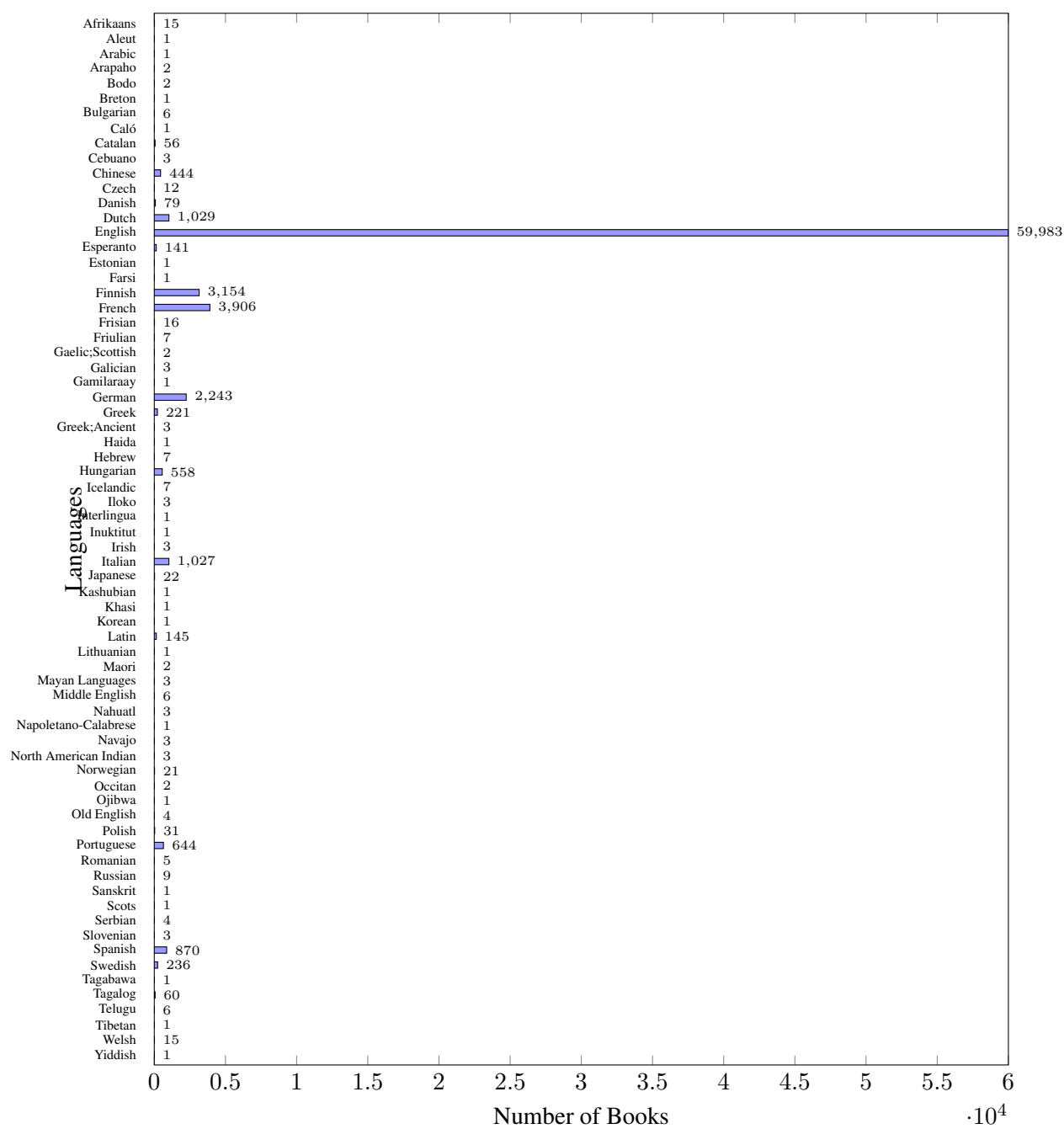


Figure 2: Total books per language in the Gutenberg Corpus

[illegible]

Table 2: The number of books added to Project Gutenberg per year per language